Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

5 solution use cases · 1 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

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Content Development & CI/CD

Write, test, lint, package and ship automation content through pipelines.

Solution use cases

Use Case: Provide standard automation developer workspaces

Description	Make the first day of a new automation developer short and the results consistent. Every developer works in the same editor setup with the same Ansible development tools and runs content in the same execution environment image that production uses, either on the laptop or in a browser-based workspace hosted on OpenShift.
Collection	`infra.automation_development_office` (Validated, v2.0.0, updated 2026-10-06), `infra.ee_utilities` (Validated, v4.6.0, updated 2026-09-17)
Key content	• `infra.automation_development_office.ocp_devspaces` — An collection. • `infra.automation_development_office.ocp_dev_hub` — Deploy Red Hat Developer Hub (RHDH) on OpenShift via the RHDH operator. • `infra.automation_development_office.install_gitlab` — Install standalone GitLab CE/EE on RHEL via official packages. • `infra.automation_development_office.aap_build_ee` — Build a custom Ansible execution environment image. • `infra.ee_utilities.ee_builder` — Excecution enviroment builder
Notes	• Continue with Test automation content and Build CI/CD for automation content. • Image build and registration are detailed in Build and manage execution environments. • Keep the workstation baseline small. Every extra tool makes it harder to reproduce a problem. • Hosted workspaces keep source code and credentials off laptops, which helps with contractors and regulated environments.

Use Case: Test automation content

Description	Catch problems in automation before it touches production systems. Content passes style and syntax checks, role and collection tests, and finally a run in AAP against a lab inventory. The same checks run on a developer machine and in the pipeline, and a failed check blocks the merge.
Collection	`ansible.controller` (Certified, v4.8.11, updated 2026-09-30), `ansible.utils` (Certified, v6.1.1, updated 2026-09-17), `infra.aap_configuration_extended` (Validated, v4.12.1, updated 2026-10-15)

Key content	• `ansible.controller.project_update` — Update a Project in Automation Platform Controller • `ansible.controller.job_launch` — Launch an Ansible Job. • `ansible.controller.job_wait` — Wait for Automation Platform Controller job to finish. • `ansible.controller.workflow_launch` — Run a workflow in Automation Platform Controller • `ansible.utils.validate` — Validate data with provided criteria • `infra.aap_configuration_extended.aap_rules_validation` — An ansible role which audit the declared AAP configuration and validate it against a set of user-defined rules.
Notes	• Tests are only as good as the lab. For network and appliance content, use virtual devices or a shared lab with scheduled access. • Linting and Molecule come with the Ansible development tools, not from a collection; this page lists the platform parts that connect them to AAP. • Continue with Build CI/CD for automation content to chain the tests with promotion. • Event-Driven Ansible can start the integration tests when a push or pull request event arrives, in place of polling.

Use Case: Build CI/CD for automation content

Description	Treat playbooks, roles and collections like software. A merge in Git triggers the tests, updates the project in Automation Platform, and promotes the same version through development, test and production environments with approvals where needed. Nobody edits content on the controller.
Collection	`ansible.scm` (Certified, v3.2.1, updated 2026-03-31), `ansible.controller` (Certified, v4.8.11, updated 2026-09-30), `ansible.hub` (Certified, v1.1.2, updated 2026-10-01), `ansible.eda` (Certified, v2.13.0, updated 2026-07-07)
Key content	• `ansible.scm.git_retrieve` — Retrieve a repository from a distant location and make it available on the execution node • `ansible.scm.git_publish` — Publish changes from a repository available on the execution node to a distant location • `ansible.controller.project` — create, update, or destroy Automation Platform Controller projects • `ansible.controller.project_update` — Update a Project in Automation Platform Controller • `ansible.controller.job_template` — create, update, or destroy job templates. • `ansible.controller.workflow_job_template` — create, update, or destroy Automation Platform Controller workflow job templates. • `ansible.eda.webhook` — Receive events via a webhook.
Notes	• Related: Test automation content and Manage Automation Platform as code, which applies the same pipeline to the platform configuration. • Keep webhook secrets and tokens in credentials, and rotate them; see Rotate vaulted secrets and platform credentials. • Use workflow approval nodes for production gates; the approvals can be answered from an ITSM tool through the API. • For large estates, prefer Git tags over branches for production so that what ran is always identifiable.

Use Case: Publish collections to automation hub

Description	Give teams one trusted place to get automation content. Internal roles and modules are packaged as collections, published to private automation hub, reviewed, approved and then installed by pipelines, execution environment builds and developers from one source. Certified and validated content can be synced alongside it.
Collection	`ansible.hub` (Certified, v1.1.2, updated 2026-10-01), `infra.aap_configuration` (Validated, v4.10.0, updated 2026-09-15)
Key content	• `ansible.hub.ah_namespace` — create, update, or destroy Automation Hub Namespace • `ansible.hub.ah_build` — Build a collection tar • `ansible.hub.ah_collection_upload` — Upload a collection artifact to Automation Hub • `ansible.hub.ah_approval` — Approve a collection in Automation Hub • `ansible.hub.collection_repository` — Create, Update, Delete repository • `ansible.hub.collection_remote` — Configure a collection remote repository • `infra.aap_configuration.ansible_config` — An Ansible Role to create ansible.cfg files based on your Automation Hub servers
Notes	• See Build CI/CD for automation content for the pipeline, and Build and manage execution environments for the images that consume the collections. • Hub objects (namespaces, repositories, remotes, groups) can be managed with the `hub_*` roles in `infra.aap_configuration`; see Manage Automation Platform as code. • Disconnected sites need an offline sync of the collections they use. • Do not publish secrets or environment-specific variables in collections. Keep them in inventories and credentials.

Use Case: Migrate existing playbooks into collections and AAP

Description	Bring the automation that runs from laptops, cron jobs and shared servers into one governed place. Scripts and playbooks are cleaned up, split into roles and collections, stored in Git and run from Automation Platform as job templates with credentials, inventories, surveys and role-based access. The same approach applies to the logic held in other tools, such as Puppet, Chef or Salt.
Collection	`ansible.controller` (Certified, v4.8.11, updated 2026-09-30), `ansible.scm` (Certified, v3.2.1, updated 2026-03-31), `infra.aap_configuration_extended` (Validated, v4.12.1, updated 2026-10-15), `redhat.rhel_system_roles` (Certified, v2.4.2, updated 2026-10-06)

Key content	• `ansible.controller.project` — create, update, or destroy Automation Platform Controller projects • `ansible.controller.inventory` — create, update, or destroy Automation Platform Controller inventory. • `ansible.controller.credential` — create, update, or destroy Automation Platform Controller credential. • `ansible.controller.job_template` — create, update, or destroy job templates. • `ansible.controller.workflow_job_template` — create, update, or destroy Automation Platform Controller workflow job templates. • `infra.aap_configuration_extended.export_job_template_related` — Export Job Template and related objects
Notes	• Combine with Build CI/CD for automation content and Test automation content so that migrated content stays maintainable. • Use Publish collections to automation hub once shared roles exist. • No Ansible collection converts Puppet, Chef or Salt code automatically; treat the old code as documentation of intent and verify with tests. • Migrating from Tower or an older platform version is a different task: Migrate from Ansible Tower or AAP 2.4 to AAP 2.5 and later.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Source control `ansible.scm`

Certified Red Hat · version 3.2.1 · updated 2026-03-31 · 2 modules → 1 use cases

Use Case: Retrieve and publish Git repositories

Description	Retrieve a repository to the execution node and publish changes to a remote location.
Collection	`ansible.scm` (Certified, Red Hat, v3.2.1, updated 2026-03-31)
Key content	**Retrieve repository** `git_retrieve` **Publish changes** `git_publish`